

Leveraging Emotion Recognition for Enhancing Arts Education: A Classroom Behavior Analysis Approach

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Significant developments in modern teaching have led to the integration of advanced equipment and technologies, including the introduction of artificial intelligence (AI) solutions. This paper proposes the utilization of AI technology to improve the teaching effectiveness of art courses through a classroom behavior analysis approach. By employing image acquisition tools and analytical software, the proposed method enables real-time data capture and comprehensive analysis of students' emotional states, thus providing valuable insights to educators for tailored pedagogical outcomes. The developed emotion recognition model, ERAM, exhibits a rapid response rate and dependable accuracy, positioning it as a robust tool for immediate feedback and adaptive teaching strategies. Experimentation revealed an 11.06% improvement in teaching effectiveness with the implementation of intelligent systems, signifying its potential to enhance learning outcomes and student satisfaction in arts education.

CCS CONCEPTS • Applied computing • Education • Computer-assisted instruction

Additional Keywords and Phrases: Smart education, Emotion recognition, Art education, Behavior analysis

1 INTRODUCTION

Arts courses (e.g., music, film, theater, art, dance, and other arts areas) generally focus on the aesthetic experience of the learning process, and the higher the level of student engagement, the more effective the instruction. Uhrmacher[1] made a conclusion that “Active engagement, sensory experience, connections, imagination, perceptivity, and risk taking are conditions that encourage aesthetic experiences. It is these experiences that seem to support deep engagement as well as memory retention, an increase in knowledge, student satisfaction, meaning making and creativity.” Teachers need to understand the degree of student participation, sensory experience, and the effect of imagination and creativity stimulation through teaching interactions and classroom observations to enhance students' aesthetic experience, and arts education administrators need a quantitative method to evaluate the effectiveness of arts programs. Historically, Hollywood has paid great attention to audience feedback as early as the 1950s through a set of quantitative observation system to evaluate the audience's reaction to the film, when Columbia Pictures introduced an electronic system to record the audience's reaction to the movie, telling the studio executives which passages have been recognized by the audience, and which parts of it are worth improving. The Disney Research Center has recently developed a neurological system to monitor audience reactions in the studio in real time, quantifying whether the film has achieved its intended effect and seeking to maximize audience

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satisfaction. We are inspired by this, in art courses, especially film courses through an intelligent tool to observe the degree of student participation, so that teachers in the teaching process in real time to obtain student feedback data, through a scientific and quantifiable method of real-time assessment of the teaching effect, to help teachers to adjust the teaching method in a timely manner, to enhance the aesthetic experience of the students and the satisfaction of the classroom.

Smart education, also known as technology-enabled or digital education, refers to the integration of advanced technologies and digital tools to enhance teaching and learning processes[2]. It leverages the power of technology to create interactive and personalized learning experiences, improve educational outcomes, and facilitate more efficient and effective educational practices. Smart education encompasses various applications and tools that transform traditional educational approaches.

The application of smart education can be divided into two categories, one for learners and another for teachers and teaching managers. There are many applications for learners, such as personalized learning and interactive learning resources [3]. Personalized Learning refers that smart education aimed to provide personalized learning experiences tailored to individual student needs and preferences. Adaptive learning systems use algorithms and data analytics to assess students' strengths, weaknesses, and learning styles, allowing for customized instruction and adaptive content delivery. Interactive learning resources refers to smart education that utilizes interactive digital resources, such as multimedia content, simulations, virtual reality (VR), and augmented reality (AR), to engage students in active learning. These resources help visualize complex concepts, create immersive learning environments, and enhance students' understanding and retention of information. However, there are few applications that serve teachers and teaching managers.

Teachers and teaching management are crucial in the education system for creating effective learning experiences. Effective teaching management maintains an organized environment, enhancing student engagement and focus. Teachers assess students' progress, provide feedback, and adjust teaching strategies. However, teachers have limited energy and cannot focus on every student. In art courses, teachers need to manage order, understand students' progress, and provide assistance. This study proposes a teaching assistance solution using facial emotion recognition. By presenting the results to teachers, they can assess students' learning status and intervene promptly to address difficulties.

1.1 Art course teaching

China's university art education emphasizes personalized and practical teaching. University art education in China focuses on developing students' individuality, encouraging them to showcase their individual characteristics in their creations, and guiding them to constantly improve their artistic skills during the creative process.

However, Chinese university art education is faced with challenges due to its special circumstances.

Firstly, class sizes are large. The student-to-teacher ratio in Chinese universities is often high, making it challenging for teachers to provide personalized attention and support to each student during art classes. Particularly, art education emphasizes the cultivation of students' individuality, which makes it difficult to effectively resolve individual problems and doubts.

Secondly, teaching resources are limited. Teachers need various teaching resources, such as musical equipment, art tools, and theaters, but these resources may be limited in schools and colleges. Teachers need to face the challenge of limited resources, seek solutions and innovative methods, and fully utilize limited teaching resources to enhance teaching quality.

Lastly, teaching evaluation and feedback are not timely. In the arts field, evaluation standards and methods are relatively subjective and flexible. Teachers need to accurately evaluate students' performance and provide constructive guidance, requiring them to possess rich professional experience and keen artistic appreciation abilities. However, it is difficult for teachers to make accurate evaluations during classroom teaching.

It is important to note that these difficulties are not exclusive to Chinese universities and can be relevant in educational institutions worldwide. Universities continuously work to address these challenges by adopting innovative teaching methodologies to enhance the learning experience in arts courses.

1.2 Research content

Artificial intelligence technology is currently advancing rapidly and has been applied in many fields such as transportation, healthcare, and security. We propose a solution that combines privacy-protected emotion recognition and operation recognition. This solution applies AI to analyze student behavior in the laboratory teaching of arts courses. It facilitates the visualization analysis and evaluation of students' classroom behavior and emotion during the teaching process, enabling teachers to make real-time adjustments based on the actual situation, thereby enhancing the quality of teaching.



Figure 1. Scenarios of AI applied to arts courses behavior analysis.

The usage scenario for applying artificial intelligence to computer laboratory behavior analysis is illustrated in Figure 1. The camera captures real-time images, which are then transmitted to the back end server. The AI model in the server utilizes facial emotion recognition to assess the learning status of each student and aggregates the results. These statistics are then sent to the teacher's computer according to the designated time intervals, for example, every 5 minutes. By analyzing these statistics, the teacher can understand the current learning status of the students and make timely adjustments to enhance the quality of teaching. For instance, in music teaching, the teacher makes students experience the emotions through an upbeat piece of music. If the emotion recognition results do not meet the teacher's expectations, it means that the students may not have seriously experienced the emotions of the piece and the teaching is not effective. At this point, the teacher can adjust the teaching method or re-interpretation to improve students' participation in the aesthetic experience of art.

2 MATERIALS AND METHODS

2.1 Solution Structure

In the realm of computational laboratory behavioral analysis, there is a significant demand for Artificial Intelligence-based solutions that uphold user privacy and offer rapid response times. To cater to these unique application prerequisites, we present a novel proposition, referred to as the Intelligent Engineering Teaching System (IEDS). The specific components and functionalities of this solution are depicted in Figure 2.

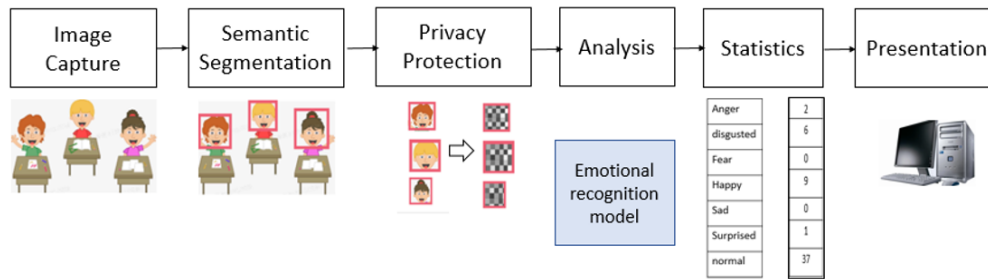


Figure 2. Diagram of solution structure.

1. The cameras capture computer laboratory image data.
2. Images captured in the computer laboratory typically contain multiple students, which necessitate a preliminary step before direct analytical processing can occur. Specifically, individual faces within these images must be partitioned into separate images, thereby preparing them for subsequent AI model analysis. The dimensions of images post-semantic segmentation are variable, and to standardize these for model training, we resize all images to a uniform size of $224 \times 224 \times 3$ pixels.
3. The incorporation of a privacy protection module assures the provision of biometric privacy. Utilizing the Fourier transform, we effectively eliminate the low-frequency information within images, converting them into high-frequency representations. These transformed images, while unrecognizable to the human eye, do not interfere with the AI model's training efficacy.
4. Analysis module. The analysis module consists of an emotion recognition model and a clustering model. The emotion recognition model identifies the processed images with privacy protection.
5. Upon the conclusion of the individual image analysis post-segmentation, we count the number of images based on unique emotional features. This ensures the safeguarding of student behavioral privacy, as their identities remain undisclosed.
6. The server sends the statistical results to the teacher's working PC.

2.2 Description of main function modules Description

2.2.1 Privacy protection

The privacy protection module processes the image in a form that cannot be recognized by the naked eye, but it does not affect the image being trained and recognized by AI models.



Figure 3. Image privacy protection processing.

As illustrated in Figure 3, low-frequency images appear more akin to the original images compared to high-frequency ones. This can be attributed to human sensitivity towards low-frequency images being markedly higher.

Consequently, high-frequency images bolster model training, while simultaneously enhancing the challenge of unaided human recognition. Through this methodology, we can proficiently safeguard the privacy of the images without compromising the quality of image recognition.

2.2.2 Emotional recognition model

This study proposes a high-performance vision model that can respond quickly, and has high accuracy. We named the model as ERAM (Emotional Recognition with Attention Mechanism). The structure of the ERAM model is shown in Figure 4.

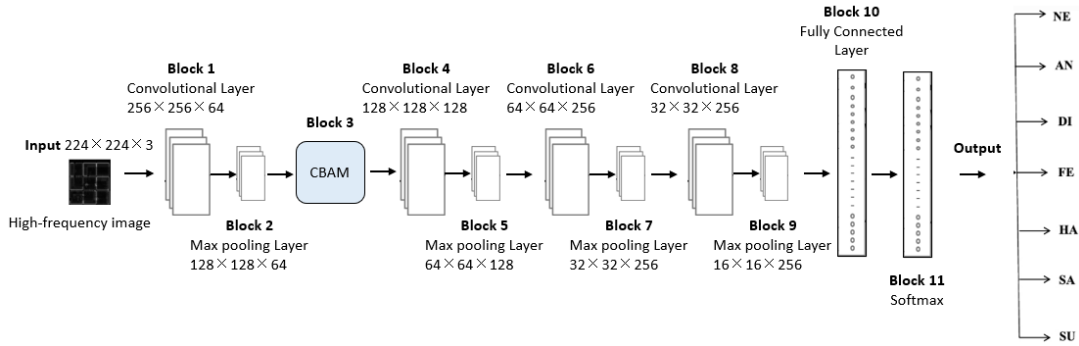


Figure 4. Structure of ERAM model.

In the privacy protection module, the collected images are divided into $224 \times 224 \times 3$ dimensions according to face. Some public sentiment recognition dataset are of this size. Our input uses the $224 \times 224 \times 3$ size to facilitate training with a public dataset. Convolution is the most common operation in neural networks. Block 1 is a convolutional layer that extracts features from input data.

Block 2 is a pooling layer that reduces the spatial dimensions of the previous layer's feature map while retaining the most important information. We use max-pooling, which is the maximum for feature points in the neighborhood. Max-pooling can retain texture features well, reduce the dimension of data, improve computational efficiency, improve the robustness of the model, and prevent overfitting.

The CBAM is a versatile and lightweight module that can be effortlessly incorporated into any neural network structure with minimal additional computational burden. Typically, the number of CBAM parameters is around 0.1 million, enabling it to enhance the precision of lightweight models without expanding the model's size. CBAM does not modify the input dimensions and can be smoothly integrated into the network model. In the context of the Emotion Recognition Attention Model (ERAM), CBAM is attached subsequent to the convolutional layer to boost the model's accuracy.

Block 4 – Block 9 are three sets of convolutional layers and max-pooling layers, with the same effect as block1 and block2, but with different filter sizes.

Block 10 is a fully connected layer. This layer effectively converts the two-dimensional feature map, produced by the convolutional output, into a one-dimensional vector. The output generated by the fully connected layer is a set of highly refined features that are then relayed to the final classification or regression process.

The output layer employs the softmax function. In instances of multi-classification challenges, the softmax function serves as the neural network output layer's activation function, enabling the model to produce probability values corresponding to the sample's potential category affiliations.

3 EXPERIMENT

3.1 Dataset

Three publicly available datasets for emotion recognition, FERPlus, CK+, and ExpW, were used in this experiment. FERPlus represents a relabeling of the original FER-2013 dataset. The training set encompasses 28,709 samples, while the testing set includes 3589 samples. The CK+ dataset, which was released in 2010, is an extension of the Cohn Kanade (CK) dataset. The dataset contains 327 image sequences. The Expression in-the-Wild (ExpW) dataset, specifically designed for facial expression recognition, incorporates 91,793 manually labeled faces with expressions. Each facial image is annotated as pertaining to one of the seven fundamental expression categories.

3.2 Performance of model

To compare the effectiveness of the ERAM model, we used some emotion recognition models and tested them using the same datasets.

VGG16[4] is a convolutional neural network model proposed by Simonyan et al., and it achieved excellent results in the 2014 ImageNet Image Classification and Localization Challenge. However, VGG16 is not a specialized model for emotion recognition. We also found some recent achievements in the research paper on emotion recognition to add for comparison. The test results are shown in Table 1.

Table 1. Comparison of multiple models.

Model	FERPlus Accuracy	CK+ Accuracy	ExpW Accuracy	Response time/k
ResNet50(transfer learning)[5]	77.67%	87.42%	74.57%	3.2518 s
VGG16	83.76%	88.57%	78.25%	3.5387 s
LER[6]	85.67%	93.45%	81.54%	2.4512s
ConvNets[7]	85.10%	92.57%	79.57%	2.2405 s
TFE [8]	84.29%	90.38%	80.95%	2.0971s
ERAM	87.64%	94.46%	82.38%	1.8617 s

Note: The model response speed is the time for the model to recognize 1000 images.

The accuracy of ERAM model reached 87.64% in the FERPlus dataset, 94.46% in the CK+ dataset, and 82.38% in the ExpW dataset. By comparison, we can see that the ERAM model has the highest accuracy in the three datasets. In addition, in terms of response time, the ERAM model is the least and it is the fastest. Among the models participating in the comparison, the ERAM model has the highest accuracy, the fastest speed, and the best performance.

4 DISCUSSION

4.1 Technological Analysis

Our experiments demonstrate that the proposed ERAM model, incorporating the CBAM attention module, outperforms other emotion recognition models in terms of accuracy and speed. Specifically designed for the lab teaching environment in engineering design courses, we have also integrated a privacy module into the model. This privacy module possesses distinctive characteristics.

For image datasets, it is essential to include both semantic (low-frequency) information and high-frequency information, although the proportion may vary. In datasets with consistent category labels across different scenes, the semantic distributions within each scene are expected to be nearly identical, assuming the same categorization. In contrast, the high-frequency distribution may contain category-specific information as well as extraneous noise that can impact model training and generalizability. Human annotators, who primarily focus on semantics, tend to overlook high-frequency information.

During the early stages of training, CNNs predominantly utilize semantic information. As training progresses and the loss plateaus, additional high-frequency components are introduced to further improve performance. Therefore, incorporating high-frequency images directly does not detrimentally affect accuracy.

4.2 Analysis of Teaching Applications

To gauge the real-world efficacy of the IEDS, we enlisted the participation of four instructors and 167 students. The student cohort was distributed among eight distinct classes. Half of these classes, during the arts course, were instructed with the utilization of the IEDS system, whereas the remaining classes were conducted traditionally. This experimental arrangement spanned one month. Subsequently, all participating trainees were evaluated at the culmination of the arts course. We utilized examination scores as a benchmark for pedagogical effectiveness. The associated distribution of educators and students is detailed in Table 2.

Table 2. The corresponding distribution of teachers and students.

	With IEDS system	Without IEDS system
Teacher A	Class1	Class2
Teacher B	Class3	Class4
Teacher C	Class5	Class6
Teacher D	Class7	Class8

The primary users of the IEDS system are instructors, and to account for the effect of diverse teaching methodologies, we solicited teachers varying in both teaching experience and gender, as delineated in Table 3.

Table 3. Teacher's personal information.

	Gender	Teaching ages
Teacher A	female	1.5 years
Teacher B	male	1years
Teacher C	female	8 years
Teacher D	male	10.5 years

Upon the conclusion of the engineering design course, we administered examinations to all students across the eight classes, with the results tabulated in Table 4.

Table 4. Engineering design course exam scores.

	With IEDS system				Without IEDS system			
	Class1	Class3	Class5	Class7	Class2	Class4	Class6	Class8
Average score	82.68	81.74	84.36	87.56	73.15	75.78	77.27	76.63
Average of group	84.09				75.71			

In terms of test results, the classes that used IEDS system in teaching had an average score of 84.09 points. The average score of the classes that did not use IEDS system was 75.71 points. Classes that used IEDS system scored 8.44% higher on exams than classes that didn't. The test results reflect the teaching effect, indicating that the teaching effect increased by 11.06% after using IEDS system.

Operating as an adjunct to manual observations, the IEDS system furnishes statistical data about students' emotional states and operational frequency. This allows teachers to tailor their instructional strategies based on student responses, thereby enhancing overall teaching effectiveness.

5 CONCLUSIONS

Research has proved that teachers of art courses can effectively adjust the rhythm of teaching activities and teaching methods according to students' real-time feedback, improve students' participation in the aesthetic experience of art, and effectively enhance the effect of knowledge acquisition and students' satisfaction through intelligent aids. At the same time, this intelligent observation equipment provides teaching managers with systematic classroom teaching data, which is conducive to organizing teaching research and improving the scientific and effective management.

In this manuscript, we propose the IEDS system for the student behavior analysis for teaching arts courses. This solution, catering to contemporary teaching approaches, assists educators in facilitating arts education, thereby enhancing the quality of instruction. Our method involves the analysis of students' learning states via emotion recognition and frequency analysis of operations, demonstrating both high accuracy and efficiency. We amass data for analysis and render them in real time, thereby enabling teachers to optimize the effectiveness of their engineering course instruction.

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